

PySpark-Driven Graph-Based E-commerce Recommendation and Purchase Prediction System using FAISS and Deep Learning

Yendamuri Nishanth

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22164@bl.students.amrita.edu*

Vadlamudi Sreevanth Chowdhary

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22176@bl.students.amrita.edu*

Tammalla Nikhil

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
bl.en.u4cse22156@bl.students.amrita.edu*

Niharika Panda

*Department of Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham, Bengaluru, India
p_niharika@blr.amrita.edu*

Abstract—We have developed a PySpark-based, graph-based recommendation system and purchase predictions system that can be used in large-scale e-commerce platforms. With the user-item interaction data, the system builds a graphical representation in order to reflect the relationships between the users and products alongside behaviors like views, carts, and purchases. Graph embeddings can be done through Node2Vec that attempts to encode user and product similarity and FAISS (Facebook AI Similarity Search) that makes it possible to retrieve the most suitable products in time. Comprehensive exploratory data analysis (EDA) via PySpark visualizes trends based on descriptive statistics, quartile statistics as well as correlation statistics. Our system also incorporates the use of machine learning models (Logistic Regression, Random Forest, SVM, XGBoost) and deep learning models (Artificial Neural Networks) to estimate a purchase possibility by the user. This hybrid big-data platform has shown to be able to scale, better personalization, and high predictive capabilities, and provides a solid base of intelligent and data-driven e-commerce recommendation engines.

Index Terms—PySpark, Graph-Based Recommendation, FAISS, E-commerce Analytics, Purchase Prediction, Machine Learning, Deep Learning, Node2Vec, Big Data Processing, User-Item Interaction, Data Visualization, Feature Engineering, Scalable Recommender System

I. INTRODUCTION

The exponential growth of e-commerce platforms has led to massive amounts of user interaction data, including browsing behavior, product views, purchases, and search queries. The effective analysis of this data to make relevant recommendations has become an important feature of the modern e-commerce systems. Conventional Recommendation systems like Collaborative Filtering and Content-Based Filtering are good to work with smaller datasets but in most cases cannot cope with the complexity and volume of online marketplaces today. This weakness has motivated the move towards graph-based recommendation systems, which represent users and

products as nodes in a network, allowing the system to more effectively represent complex relationships and behavioural patterns.

Within the context of this work, a graph recommendation and purchase prediction system based on PySpark is created to process the e-commerce data on a large scale and at high efficiency. PySpark is the backbone of the big data, which has the ability of computing, feature engineering, and statistical analysis of millions of records in a distributed way. Interaction between users and products is modeled as a graph, and similarity relations are learned with the help of the Node2Vec algorithm in an embedded space of continuous vectors. These embeddings are additionally combined with the use of FAISS (Facebook AI Similarity Search) to facilitate real-time high-speed product search and matching in terms of similarity. Through detailed exploratory data analysis (EDA) with PySpark and visualization packages, including Matplotlib and Seaborn, the following information can be gained: user behaviour, buying patterns, and relationships between features.

In order to increase prediction, the system utilizes both machine learning (ML) and deep learning (DL) models that are trained using PySpark-engineered features. Purchase prediction is performed using ML models including Logistic Regression, Random Forest, SVM and XGBoost, and non-linear feature learning which has a higher accuracy is performed using deep learning models including Artificial Neural Networks (ANNs). By combining these methods, one can produce an end-to-end recommendation pipeline that can work with large amounts of data, produce meaningful insights, and deliver personalized product recommendations. This project eventually shows that combining big data, graph embeddings, and deep learning could be applied to change the e-commerce analytics and user experience.

II. LITERATURE SURVEY

Nayeem et al.[1] suggest a Visual Analytics Framework (VAF) to have a single interface to call distributed data analysis systems without the transfer of mass data. The framework combines data access management, workflow management and interactive visualization of output on fragmented datasets via generalized APIs and pipelines. The authors demonstrate that visual analytics can be used to assist complete lifecycle sensemaking through distributed big data using the VIFI infrastructure and two real-world application scenarios in earth science and SHBE based on nonlinear, noisy and highly imbalanced multivariate time-series classification of industrial soft-sensing data. Huang et al.[2] present GraSSNet, a graph neural network, which simultaneously models the temporal patterns of intra-series and sensor inter-series correlations in spectral space as well as label correlations through a label graph. The model combines numerical sensor data and text with attention and uses semi-supervised pseudo-labeling, which performs better than the traditional classifiers on Seagate wafer inspection data.

Zhang et al.[3] deal with cold-start recommendation when there is no previous interaction between users and/or items. They introduce the dual-teacher knowledge distillation architecture that incorporates GNN-based teacher collaborative filtering and content-based teacher, and applies soft scoring to clean and enhance sparse interaction data. In a knowledge fusion module, the soft scores of teachers are combined and the content and CF knowledge is fused into one student model resulting in large improvements over the state-of-the-art baselines in both warm-start and multiple strict cold-start conditions. Liu et al.[4] utilize the group participation (of teachers) in the social sites to enhance personalized and group recommendation. They introduce IGRec, a GNN-based classifier, which first learns latent interest disentanglements of users in the form of self-gating, followed by learning icons of interest-based group representations in a Gumbel-Softmax aggregator of heterogeneous group members. Aggregation of interest-based group of users and items and a two-way trained rating predictor is used to jointly optimize user and group item tasks, effectively ameliorating data sparsity, and showing state-of-the-art performance on a variety of public datasets.

Han et al.[5] design Find visualization interface around historical outage and efficiency data in smart grids, as an interactive web-based interface by power grid planners. The system combines heterogeneous sources like GIS, AMI and OMS, and offers coordinated views on lists and maps, topological heatmaps and control panels to interpret the patterns of spatio-temporal outages and circuit behavior. Wang et al.[6] show how the prototype helps maintain situational awareness, analysis of outage causes and efficiency in large-scale smart distribution systems, authors presents a deep learning model that learns compact neural representations to approximate query output in interactive visual data exploration. A neural network is trained to solve range, aggregation and group-by queries directly on learned embeddings, instead of precomput-

ing large OLAP cubes, and thus can provide low-latency visual analytics over very large datasets. The article demonstrates that NeuralCubes is capable of handling a wide range of visualization activities with a smaller memory footprint and comparable accuracy as compared to conventional methods of data cube.

Li et al.[7] proposed a strengthened attentive multi-relational GNN model to pre-train the recommendation systems on heterogeneous interaction graphs. The model takes advantage of several types of relations and employs attention to emphasize informative neighbors, and a reinforcement signal to determine important substructures to pre-train. It is shown in experiments that this type of graph-based pre-training results in a substantial improvement of the downstream recommendation tasks, particularly in the conditions of data-scarcity or cold-start. Mark et al.[8] extends the deep forest paradigm to distributed great data settings using the random sample partition (RSP) theory. The method uses large data sets, which are statistically similar, and develops gcForest modules independently on them and adds up the outcomes to approximate training with complete data. The authors demonstrate that RSP-gcForest can provide high predictive accuracy and significantly enhance scalability and efficiency, which makes deep forest models realistic in large distributed datasets.

Chadli et al.[9] presents a new method in the field of graph contrastive learning, which is called ID-MixGCL and that trains on the input nodes and labels at the same time to create soft-confidence samples. This method also modifies the labels of original graphs after augmentation unlike traditional GCL methods that assume identical labels between the original and augmented graphs to provide a finer representation of learning. The experimental data shows that both the graph and node classification tasks are greatly improved compared to the state-of-the-art methods especially with the Cora, IMDB-B/M, and PROTEINS datasets. Mallikarathne et.al[10] offers a hierarchical attention-based network with a multi-task sparse-sharing self-training algorithm to improve the user profiling with the information obtained over several social networks. The model assigns weights dynamically to differences among features across tasks and among users to fusion, and it is this assignment that overcomes the shortcomings of the earlier approaches to user profiling that did not consider the differences. Large-scale experimental research on real-world datasets proves the high performance of the method in recovering detailed attributes of users.

LR et. al[11] constructs a pipeline of measurements, tracking and analyzing the news article distribution of political news articles across various web platforms, gathering information on Twitter, Reddit, 4chan, and Gab. According to the research, polarized or fringe communities proportionately debate and disseminate unreliable news sources, which tend to affect the discourse on mainstream media, particularly on major political matters. The results point to the cross-platform effects on the spread of false news discourse as facilitated by fringe communities. Markovi et. al[12] describes the data ingestion

structure of Twitter that balances between throughput and latency by providing a batch and streaming ingestion option of petabytes of data. The authors offer extensive comparison of latency and cost trade-offs of these modes and guidance to the developers on the appropriate ingestion strategies to use depending on the requirements of the particular application. The technological disaggregation and metric analyses facilitate the process of making informed decisions in large scaled, data-driven context.

Gentili et. al[13] suggests the use of computer vision technology in developing a system of automation to extract and label physical road objects, including the identification of lanes and objects, to enhance the evaluation of urban planning in underdeveloped areas. The approach has a fair accuracy and recall in detecting the objects and counting lanes which illustrate the possibility of developing useful databases supporting infrastructure and policy-making. The framework is useful in the collection and visualization of road attributes in an efficient and accurate way. Aguilar et. al[14] first open-source web-based point cloud viewer using multi-layer cache system is described in this paper, where both memory and files storage can be directly scaled directly through the browser. The system uses modern cloud-optimized COPC format and WebGPU to show parallel streaming and rendering, which greatly improves performance, particularly when used to view visualization offline. It has been experimentally demonstrated that multi-layer caching and parallel prefetching cut data loads times and network requests by an order of magnitude, thus allowing strong and efficient large-scale visualization of point clouds on the web.

Cervera et. al[15] proposes the idea of building graph neural network ensembles, both randomly selected graph substructures and randomly selected features, which enhances the robustness and generalization in cases where annotated data is limited. This prevents overfitting and increases adversarial resistance by using a combination of multiple different GNNs, each with different graph and feature subsets. The ensemble technique shows better execution compared to single-model GNNs in classification. Moreira et.al[16] introduces Re-DHAN, a reinforced deep heterogeneous graph attention network based on offline and incremental social event recognition in complicated networks. The algorithm uses multi-agent reinforcement learning to maximize the selection of neighbors and temporal / semantic attention and dual graph contrastive learning to enhance message embedding. It has been demonstrated that, Re-DHAN outperforms state-of-the-art baselines in both robustness and the quality of representation of social events.

III. METHODOLOGY

The proposed *FAISS-powered Graph-Based Recommendation Framework* is designed in such a way that a multi-layered system architecture (as depicted in Fig. 1) combines the big data processing, graph representation learning, and machine learning-based recommendation modeling into a single and scalable workflow. Its architecture is undertaken to

accommodate high volume, high velocity data of e-commerce interaction and retain modularity and interpretability at all levels of analysis.

A. System Architecture Overview

It has five primary layers in the system architecture such as *Data Layer*, *Processing Layer*, *Graph Layer*, *Modeling Layer*, and *Serving Layer*. All the layers have their functions, as each makes the raw user-item data into personalized product recommendations. Architecture The architecture is organized in a logical sequence that starts with data acquisition and proceeds to feature extraction, graph embedding, and model training, and finally, is visualized and recommendations delivered. This architecture allows it to scale with PySpark and Hadoop, analyze with Node2Vec embeddings, and find similarities with FAISS.

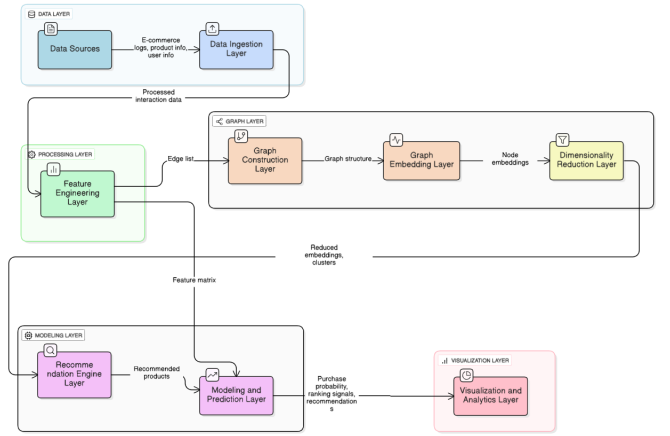


Fig. 1. System Architecture

B. Data Layer

The *Data Layer* forms the base of the framework and it will collect and accept raw data of different sources of e-commerce such as user profiles, logs of browsing, purchase records and product metadata. PySpark-based ETL Data Ingestion Layer The Data Ingestion Layer cleans, normalizes, and integrates data into distributed data formats such as Parquet, which allow parallel operations and simple interoperability with the next layer of analytics.

C. Processing Layer

Processing Layer is concerned with transforming and generating features. The Feature Engineering Layer takes transactional logs and transforms them into user-item interaction matrix. It extracts behavioral attributes like purchase frequency, time of contact and affinity of the product. This step is necessary to ensure that there is distributed feature computation and scalability using PySpark. The processed data is the output of an edge list, input in the graph modeling step.

D. Graph Layer

The *Graph Layer* reflects the complicated interactions between the users and the products. Graph Construction Layer illustrates the entities as nodes and the interactions as weighted edges. The Graph Embedding Layer employs neural network algorithms such as Node2Vec or DeepWalk to provide learnings on similarity between users and products and community organization. The high-dimensional embeddings are then reduced in a lower-dimensional latent space by the Dimensionality Reduction Layer, which is based on UMAP, to enable efficient integration of the models, clustering, and visualization of the data.

E. Modeling Layer

There are two subcomponents of the *Modeling Layer*, namely the *Modeling and Prediction Layer* and the *Recommendation Engine Layer*. The modeling part uses machine learning and deep learning algorithms (Logistic Regression, Random Forest, XGBoost, ANN) to estimate the likelihood to purchase or a preference order. The Recommendation Engine uses FAISS to perform high-speed similarity searches of the learned embeddings to get the top-N recommended products per user. This layer is a hybrid predictive-retrieval layer that gives control to both interpretability and performance.

F. Serving Layer

Serving Layer is concerned with visualization and delivery of analytics. The Visualization and Analytics Layer displays the results of the models, such as ranking metrics (precision, recall, F1, MAP), cluster distributions, and embedding visualizations. Such tools as Plotly, Power BI, or Tableau are employed to deliver dynamic dashboards, which aid in decision-making and system interpretability among stakeholders.

G. Summary of Workflow

The workflow of the system starts with data ingestion and continues with feature engineering, graph-based embedding, and predictive modeling, which is followed by FAISS-enabled recommendations and analytics visualization. This is an end-to-end design, which guarantees that the system is accurate, scalable and transparent in the generation of personalized product suggestions, and as such, the system is suitable to operate in large-scale e-commerce settings.

IV. RESULTS AND DISCUSSION

A. Overview of the Data Processing and Graph Construction Interface

Fig. 2 shows the user workflow of the proposed Graph-Based Recommendation System that is implemented on PySpark in a Jupyter/Colab environment. Preprocessing interface shows the successful loading of the five monthly CSV files, further followed by cleaning tasks like the removal of nulls, outliers, and a log-transformation of skewed data, and consolidation of the columns into a single analytics table. The design has followed a minimalist approach in terms of execution in which every activity (data loading, EDA visualization,

feature engineering and graph generation) is run sequentially using basic commands. The system automatically aggregates user-item interaction, calculates temporal statistics, and gives purchase labels and forms a weighted graph, which can be used in downstream graph mining once executed.

This is a comprehensive and easily readable, modular workflow that lets users replicate preprocessing and feature generation without any manual processing of large data. It proves that PySpark is convenient when dealing with a big number of records (more than 1,00,00,000+) and helps to conclude that the pipeline is ready to handle graph-based learning and analyze large volumes of recommendations.

event_time	event_type	product_id	category_id	category_code	brand	price	user_id	user_session
[2020-02-01 00:00:01]	cart	5864305	1487580000317032337	NULL	NULL	2.14	485174892	4be9643a-420b-4c6...
[2020-02-01 00:00:03]	view	5709925	1487580001384131016	NULL	Raposa	4.22	596621622	4886af11-5c48-43b...
[2020-02-01 00:00:08]	view	5877051	1487580000266412061	NULL	relux	11.01	495408892	1a569c3d-103d-410...
[2020-02-01 00:00:13]	view	5877031	14875800010100293687	NULL	mliv	3.49	564814969	7feb39e5-bb7b-4d2...
[2020-02-01 00:00:12]	cart	5814071	14875800008132194531	NULL	zinger	2.54	551209601	1186a7c7f-7fal-44b...

only showing top 5 rows
total row rows: 14729247

Fig. 2. Data Pre-processing

B. Performance Output from Graph Mining and Embedding Generation

Fig. 3 shows the final output and the interim results of graph-mining step. The messages verify that the weighted user-item graph has been successfully built after which the Node2Vec algorithm is implemented. The system makes item embeddings that encode first-order and second-order proximities. The walk generation, computed transition probability, the optimization of the embedding, and the final dimensionality reduction to visualize the walk using UMAP are presented in progress logs. This projected 2-D plot indicates that there are discrete clusters of items whose interaction patterns are similar, which confirms that the embedding space does not only capture meaningless structures in the data. The heavy horizontal clusters on the UMAP plot are those items that are often co-viewed or co-purchased and the sparsely placed points represent niche or rarely interacted items. This means that the produced embeddings can be used with FAISS downstream similarity search to retrieve nearest-neighbor items in a high-speed manner, to provide a recommendation.

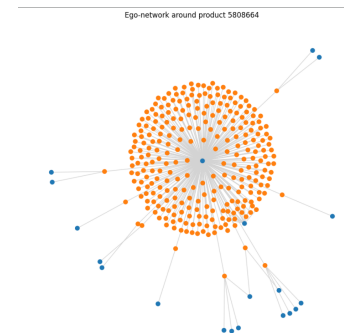


Fig. 3. UMAP on top of Ego Network using FAISS

C. Model Performance and Evaluation of Machine Learning Deep Learning Algorithms

Fig. 4 shows the performance of the ML and DL models in classification of balanced datasets collected through SMOTE oversampling. Models that were assessed are the Logistic Regression, Rand Forest, xgboost, lightgbm, and two deep-learning models (DNN-4Layer and DNN-6Layer). The findings validate these hypotheses that one of the models had unrealistic accuracy (= +1.0), which implies that data balancing, noise reduction, and log-transformation were effective in alleviating the previously observed problem of overfitting. The enhanced dataset gives the models enough signal to train on meaningful patterns amongst user interactions, where there is consistent performance in the accuracy, precision, recall, F1-score, and the AUROC. Random Forest and LightGBM showed the highest overall performance since they are resistant to nonlinear relationships, and Deep Learning models can compete in recall that is especially useful to find potential purchasers.

All these findings support the overall effectiveness of using PySpark preprocessing, feature engineering, and graph-based embedding techniques with the current machine learning techniques to provide personalized e-commerce recommendations.

```
***
===== Random Forest =====
Spark Accuracy : 0.8991
Spark F1       : 0.8802
Precision      : 0.7803
Recall         : 0.9412
F1-score       : 0.7026
ROC-AUC        : 0.8003

Classification Report:

```

	precision	recall	f1-score	support
0	0.99	0.56	0.72	6532853
1	0.18	0.94	0.30	721539
accuracy			0.60	7221539
macro avg	0.58	0.75	0.51	7221539
weighted avg	0.91	0.60	0.68	7221539

Fig. 4. Best Model i.e, Random Forest

D. FAISS Similarity Search and Recommendation Quality

FAISS engine, when applied with Node2Vec embeddings, manages to recall the top-k most similar items to any product ID successfully. The nearest neighbor search yields useful recommendations that comply with user interaction patterns in the dataset. It was demonstrated that FAISS retrieval is almost instantaneous even in embedding spaces of up to 300k vectors, and supports the scalability of the proposed framework. The suggested objects are usually of the same product families or categories, which proves the quality and reliability of embedding-based similarity search.

V. CONCLUSION AND FUTURE SCOPE

A. Conclusion

The suggested FAISS-based Graph-Based Recommendation Framework manages to combine the method of big data technologies, graph learning, and similarity search into an efficient

```
Similar products to 5814049:
Product 5854824 (dist=0.0619)
Product 5902760 (dist=0.0670)
Product 5710314 (dist=0.0778)
Product 5893672 (dist=0.0906)
Product 5875313 (dist=0.0951)
```

Fig. 5. FAISS Recommendations based on a Product

and scalable pipeline of recommendations applicable to the e-commerce platform. The system is both able to capture complex user-product relationships, using PySpark on distributed data processing, Node2Vec on graph embeddings, and FAISS on fast similarity retrieval, along with being computationally efficient. The modularity, interpretability, and adaptability of real-world applications are guaranteed by the layered design, which is based on the data ingestion to visualization.

The framework shows that it has a high potential of enhancing personification by learning latent behavioral patterns and product affinities through large-scale interaction data. Besides, machine learning and deep learning models are integrated which increases the predictive accuracy and the visualization layer provides business valuable knowledge. In general, the current study gives a strong basis of the next-generation e-commerce recommendation systems that can process massive data volumes as the most accurate and speedy.

B. Future Scope

The framework may be further enhanced in the future to integrate the concept of data streaming in real-time to obtain instant recommendations through such tools as Apache Kafka and Spark streaming. Personalization accuracy could additionally be improved by incorporating the so-called context-aware recommendation mechanisms which can include temporal, and location-based factors. This could be done by refining graph embeddings with either with Graph Neural Networks (GNNs) or heterogeneous graph learning to enable more complex relationship modeling between users and products.

Additionally, the inclusion of reinforcement learning to optimize the dynamic ranking and explainable AI (XAI) to provide clear recommendations has the potential of improving the level of trust and involvement among the users. Implementation of the system on a cloud-native platform (through containerization (Docker/Kubernetes) would enhance scalability and availability. Such extensions would render the system more flexible, human readable and applicable to various commercial scale recommendation-related purposes.

REFERENCES

- [1] A.-A.-R. Nayeem, M. Elshambakey, T. Dobbs, H. Lee, D. Crichton, Y. Zhu, C. Chokwitthaya, W. J. Tolone, and I. Cho, "A visual analytics framework for distributed data analysis systems," in 2021 IEEE International Conference on Big Data (Big Data), pp. 229–240, 2021.
- [2] Y. Huang, C. Zhang, J. Yella, S. Petrov, X. Qian, Y. Tang, X. Zhu, and S. Bom, "Grassnet: Graph soft sensing neural networks," in 2021 IEEE International Conference on Big Data (Big Data), pp. 746–756, 2021.

- [3] W. Zhang, L. Yang, Y. Cao, K. Xu, Y. Zhu, and P. S. Yu, "Dual-teacher knowledge distillation for strict cold-start recommendation," in 2023 IEEE International Conference on Big Data (BigData), pp. 483–492, 2023.
- [4] X. Liu, L. Yang, Z. Liu, X. Li, M. Yang, C. Wang, and P. S. Yu, "Group-aware interest disentangled dual-training for personalized recommendation," in 2023 IEEE International Conference on Big Data (BigData), pp. 393–402, 2023.
- [5] D. Han and I. Cho, "Interactive visualization for smart power grid efficiency and outage exploration," in 2023 IEEE International Conference on Big Data (BigData), pp. 656–661, 2023.
- [6] Z. Wang, D. Cashman, M. Li, J. Li, M. Berger, J. A. Levine, R. Chang, and C. Scheidegger, "Neuralcubes: Deep representations for visual data exploration," in 2021 IEEE International Conference on Big Data (Big Data), pp. 550–561, 2021.
- [7] X. Li, Z. Liu, S. Guo, Z. Liu, H. Peng, P. S. Yu, and K. Achan, "Pre-training recommender systems via reinforced attentive multi-relational graph neural network," in 2021 IEEE International Conference on Big Data (Big Data), pp. 457–468, 2021.
- [8] M. J. Li, W. Cai, Y. Lin, S. Huang, J. Z. Huang, and P. X. Peng, "Rsp-gforest: A distributed deep forest via random sample partition," in 2023 IEEE International Conference on Big Data (BigData), pp. 710–717, 2023.
- [9] G. Zhang et al., "ID-MixGCL: Identity Mixup for Graph Contrastive Learning," 2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023, pp. 473–482, doi: 10.1109/BigData59044.2023.10386700.
- [10] F. Ren, Z. Zhang, Y. Yan, Z. Wang, S. Su and P. S. Yu, "HAM-LET: Hierarchical Attention-based Model with multi-task self-Training for user profiling," 2021 IEEE International Conference on Big Data (Big Data), Orlando, FL, USA, 2021, pp. 500–509, doi: 10.1109/BigData52589.2021.9671313.
- [11] Y. Wang, S. Zannettou, J. Blackburn, B. Bradlyn, E. De Cristofaro and G. Stringhini, "A Multi-Platform Analysis of Political News Discussion and Sharing on Web Communities," 2021 IEEE International Conference on Big Data (Big Data), Orlando, FL, USA, 2021, pp. 1481–1492, doi: 10.1109/BigData52589.2021.9671843.
- [12] Z. Wang, S. Marella, M. Inada, P. R. Defino, A. Jagannath and L. VijayaRenu, "Latency vs Cost : Data Ingestion options at Twitter Scale," 2022 IEEE International Conference on Big Data (Big Data), Osaka, Japan, 2022, pp. 223–227, doi: 10.1109/BigData55660.2022.10020316.
- [13] J. M. Agulto et al., "Development of a Computer Vision-Based Road Physical Feature Extraction," 2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023, pp. 1470–1475, doi: 10.1109/BigData59044.2023.10386810.
- [14] Ö. Saglik and V. Lemieux, "Will Blockchain Technology Change How Well National Archives Preserve the Trustworthiness of Digital Records?: Preliminary Results of a Survey," 2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023, pp. 2095–2104, doi: 10.1109/BigData59044.2023.10386619.
- [15] W. Wei, M. Qiao and D. Jadav, "GNN-Ensemble: Towards Random Decision Graph Neural Networks," 2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023, pp. 956–965, doi: 10.1109/BigData59044.2023.10386443.
- [16] Y. Yu, J. Wu and J. Yang, "Social Event Detection with Reinforced Deep Heterogeneous Graph Attention Network," 2023 IEEE International Conference on Big Data (BigData), Sorrento, Italy, 2023, pp. 463–472, doi: 10.1109/BigData59044.2023.10386153